

Erdős Problem #354

EP-354

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $\alpha, \beta \in \mathbb{R}_{>0}$ such that α/β is irrational. Is the multiset $\{\lfloor \alpha \rfloor, \lfloor 2\alpha \rfloor, \lfloor 4\alpha \rfloor, \dots\} \cup \{\lfloor \beta \rfloor, \lfloor 2\beta \rfloor, \lfloor 4\beta \rfloor, \dots\}$ complete? That is, can all sufficiently large natural numbers n be written as $n = \sum_{s \in S} \lfloor 2^s \alpha \rfloor + \sum_{t \in T} \lfloor 2^t \beta \rfloor$ for some finite $S, T \subset \mathbb{N}$?

What if 2 is replaced by some $\gamma \in (1, 2)$?